

How to Model Defects and Surfaces

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9th September 2025

"Like human defects, those of crystals come in a seemingly endless variety, many dreary and depressing, and a few fascinating."

by Ashcroft/Mermin - SOLID STATE PHYSICS

Outline

DEFECTS and ... SURFACES

G. Mallia

Perfect vs
Real Crystals

Defect
Extension

Point Defects
Classification

Point Defects
Notation

Approaches

Running
Simulations

SURFACES

- 1 Perfect vs Real Crystals
- 2 Defect Extension
- 3 Point Defects Classification
 - Point Defects in Bulk (3D)
 - Point Defects in Nanoscale Materials (2D, 1D and 0D)
- 4 Point Defects Notation
- 5 Approaches
 - Non Periodic Approaches
 - Periodic Approaches
- 6 Running Simulations
- 7 SURFACES

Array of Objects

DEFECTS and ... SURFACES

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by Rachele Fermani

Perfect vs Real Crystals

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Perfect Crystal :

3D periodic array of atoms
short range order
long range order

Real Crystal :

presence of defects
disorder
entropic contribution
limited in size

$T=0K \rightarrow$ PERFECT CRYSTAL

Defect Extension

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0D :

■ POINT DEFECTS

1D - Line Defects :

■ Dislocations (Edge and Screw)

2D - Plane Defects :

■ Surfaces

■ Boundaries / Microcrack : Grain, Tilt and Twin

■ Interfaces

■ Stacking fault (i.e. ABCABC, ABC**AC**BABC)

■ Magneli phases

3D - Volume Defects :

■ Voids

Point Defects in Bulk (3D)

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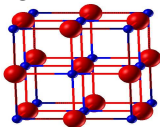
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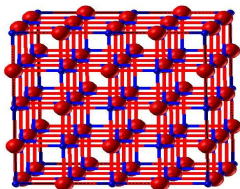
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PERFECT CRYSTAL

MgO conventional cell



MgO 2x2x2 supercell



Point Defects in Bulk (3D)

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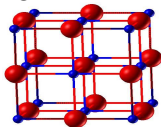
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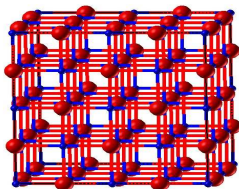
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PERFECT CRYSTAL

MgO conventional cell

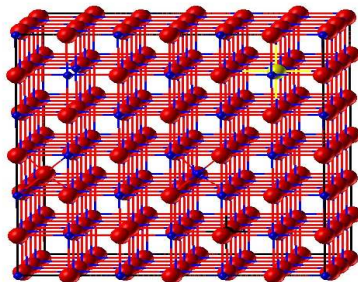


MgO 2x2x2 supercell



DEFECTIVE CRYSTAL

MgO 3x3x3 supercell



Point Defects in Bulk (3D)

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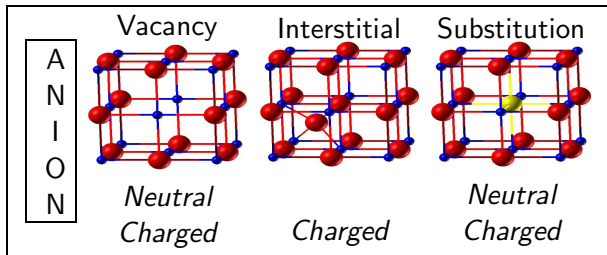
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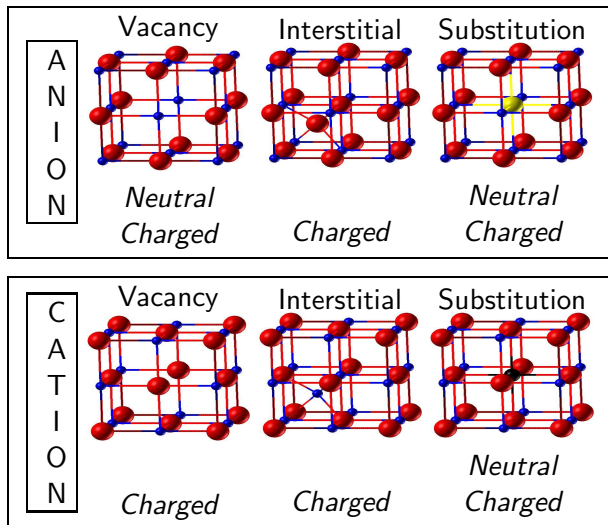
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NOTE: Self-interstitial atom!!!

Simple vs Complex Defects

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- Simple: only one atom is involved
- Complex: more than one atom, interaction between simple defects
 - Schottky
 - Frenkel

Point Defects in Compounds: Schottky Defects

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- in compounds with formula MX
the number of cation vacancies is equal to the number of anion vacancies;
- MX_2
twice as many anion vacancies as cation vacancies;
- M_2X_3
two vacancies on the cation sublattice and three vacancies on the anion sublattice;

1 NEUTRALITY

2 NO CLUSTERING OF THE VACANCIES IS REQUIRED

Point Defects in Compounds: Frenkel Defects

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An atom from one sub-lattice moves to a normally empty place in the crystal, leaving a vacancy behind.

The number of vacancies is not connected to the formula of the compound.

NEUTRALITY

- 1 Frenkel defect: a cation moves to an interstitial position;
- 2 anti-Frenkel defect: an anion moves to an interstitial position;

Point Defects at Surface (2D)

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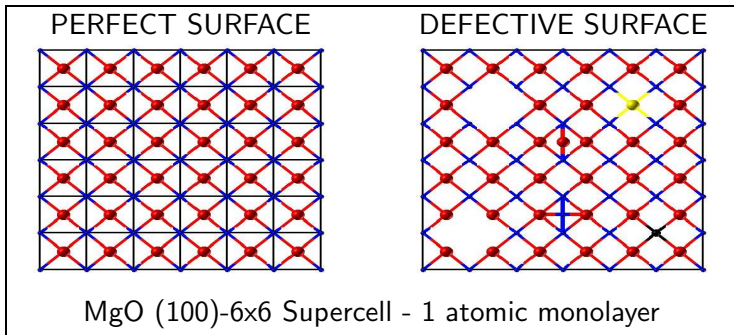
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SURFACES



- Vacancy
- Interstitial
- Substitutional

Point Defects at Surface (2D): Layer Effect

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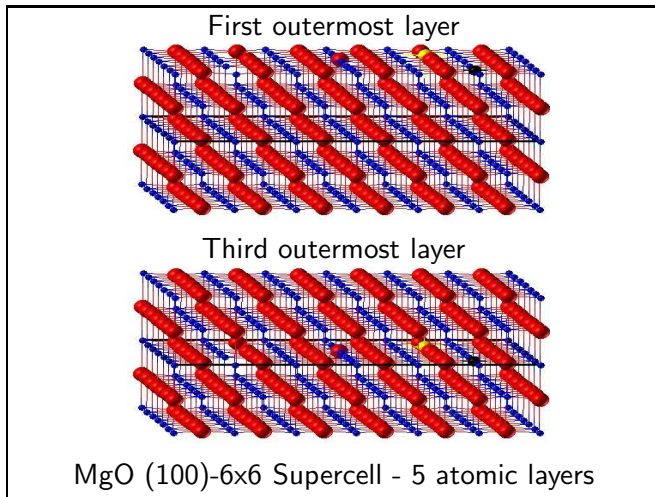
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Diffusion of point defects!!!

Point Defects in Nanoscale Materials (2D, 1D and 0D)

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SURFACES

- Thin film (2D)
- Nanotube (1D)
- Quantum dot (0D)

Crucial for application and electronic devices!!!

Point Defects in 1D: Nanotube ¹

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Perfect Nanotube :

Prevailing conception of perfect crystalline carbon nanotube and particularly single-walled nanotubes (SWNTs);

VERSUS

Real Nanotube :

High-quality SWNTs contain one defects per $4\mu\text{m}$ (1 site per 10^{12} atoms) on average, with a distribution weighted towards the areas of SWNT curvature: curved SWNT segments show higher defect density than straight segments;
SWNT defect sites are more reactive than the pristine sp^2 -bonded lattice;
(*defect density comparable with high-quality silicon single-crystals*)

¹Y. Fan et al, *Nat Mater*, 4(24):906,(2005)

Point Defect Notation (I)

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NOTATION DEPENDENT ON THE PROPERTIES UNDER INVESTIGATION ²

- the ligand-field spectrum depends on **the defect charge** (unimportant to know the host crystal)

²P. A. Cox, *Transition Metal Oxides*

Point Defect Notation (I)

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NOTATION DEPENDENT ON THE PROPERTIES UNDER INVESTIGATION ²

- the ligand-field spectrum depends on **the defect charge** (unimportant to know the host crystal)

Example: whether Cr^{3+} replaces

Mg^{2+} in MgO

Al^{3+} in Al_2O_3

Ti^{4+} in TiO_2

²P. A. Cox, *Transition Metal Oxides*

Point Defect Notation (I)

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- the type of electronic carrier in a semiconduction depends on **the imbalance charge** between the defect charge and the charge present in the same site in the perfect crystal

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Point Defect Notation (I)

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Ti^{4+} in TiO_2

- the type of electronic carrier in a semiconduction depends on **the imbalance charge** between the defect charge and the charge present in the same site in the perfect crystal
Example: (n-type versus p-type)

²P. A. Cox, *Transition Metal Oxides*

Point Defect Notation (II): The Kröger-Vink Notation ³

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- 1 the entity occupying the defect site
(M, X, V or $\square$ or substitutional elements)
 - M is the cation (or less electronegative) component in a compound crystal
 - X is the anion (or more electronegative atom): a nonmetallic element like O, N, C, etc
 - V is the vacancy; $\square$ is preferable in case of vanadium

³R. T. DeHoff, *Thermodynamics in Materials Science*

Point Defect Notation (II): The Kröger-Vink Notation ³

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- 2 a subscript for the type of site occupied (M, X or i)
 - i is an interstitial site

³R. T. DeHoff, *Thermodynamics in Materials Science*

Point Defect Notation (II): The Kröger-Vink Notation ³

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 - X is the anion (or more electronegative atom): a nonmetallic element like O, N, C, etc
 - V is the vacancy; $\square$ is preferable in case of vanadium
- 2 a subscript for the type of site occupied (M, X or i)
 - i is an interstitial site
- 3 a superscript for the excess charge associated with the site, ($'$), ($\cdot$) and ($\times$).
 - ' designates a unit of excess of negative charge
 - $\cdot$ represents an excess of positive charge
 - $\times$ no charge excess

³R. T. DeHoff, *Thermodynamics in Materials Science*

Point Defect Notation (III): The Kröger-Vink Examples

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■ NEUTRAL

O_O^x : a normal site in any oxides

Al_{Al}^x : a normal site in crystal like alumina (Al_2O_3)

Cr_{Al}^x : a trivalent chromium ion on a cation site in Al_2O_3

■ POSITIVE

$h^\cdot$: electron hole

$Al_i^{\cdot\cdot\cdot}$: a trivalent aluminium ion in an interstitial site;
its three positive charges are all excess charge

Ca_K : a divalent calcium ion on a normally monovalent
potassium site in KCl

$V_O^\cdot$: an oxygen ion vacancy ($\square_O^\cdot$)

■ NEGATIVE

$e'^\cdot$: free (unattached) electron

O_i'' : an oxygen ion in an interstitial site;

Mg'_{Al} : a divalent magnesium ion on a normally trivalent
aluminium cation site;

V_{Al}''' : an aluminium ion vacancy ($\square_{Al}'''$)



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APPROACHES

Non Periodic Approaches (I): CLUSTER

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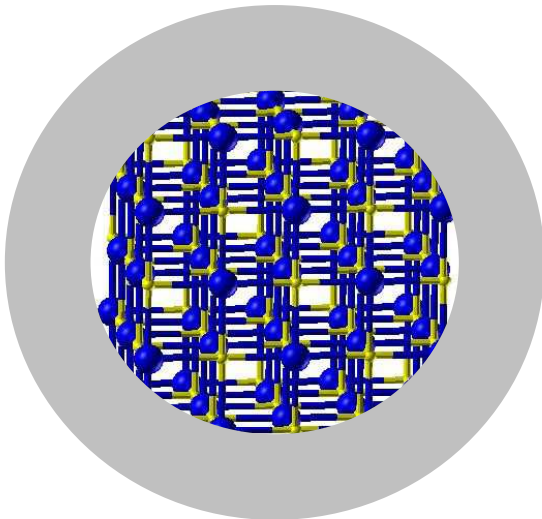
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Non Periodic Approaches (II)

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SURFACES

- BARE CLUSTER
- DRESSED and SATURATED CLUSTER
(covalent crystals)
- ONIOM: Multi level technique MM/QM
MODEL CLUSTER and REAL SYSTEM

$$E_{tot} = E_{cluster}^{high} + \Delta E$$

$$\Delta E = E_{system}^{low} - E_{cluster}^{low}$$

PROS

- High level computational method

CONS

- size and shape
- boundaries

Periodic Approaches

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SURFACES

- **SUPERCCELL (SIZE CONSISTENCY)**
- P-ONIOM: Multi level technique MM/QM
MODEL CLUSTER and PERIODIC REAL SYSTEM
- Embedding

PROS

- No boundaries problem

CONS

- size and shape

Supercell approach

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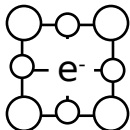
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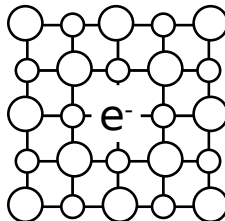
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EXAMPLE: F – center in LiF



- Concentrations
- Isolated limit

Supercell approach: Size consistency

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n	Lattice	E_p^n	E_p^2
8	P	-428.221800	-107.055450
16	F	-856.443601	-107.055450
32	I	-1712.887207	-107.055450
54	F	-2890.497155	-107.055450
64	P	-3425.774381	-107.055449
128	F	-6851.548839	-107.055451
216	P	-11561.988593	-107.055450
250	F	-13381.931279	-107.055450
256	I	-13703.097661	-107.055450

Embedding

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- to subdivide the entire defect system into
 - 1 a molecular cluster (C) containing the defect, and
 - 2 an external region (D), the indented crystal
- to calculate the wave function for the molecular cluster in the field of the indented crystal
- to correct the cluster solution in order to allow for the propagation of the wave function into the indented crystal while generating the density matrix of the defect system.



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RUNNING SIMULATIONS

Running Simulations: Keywords - SUPERCEL (I)

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SUPERCEL: to expand the primitive cell

```
SUPERCEL
2 0 0
0 2 0
0 0 2
ENDG
```

SUPERCON: to expand the conventional cell

```
SUPERCON
2 0 0
0 2 0
0 0 2
ENDG
```

Running Simulations: Keywords - SUPERCEL (II)

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The expansion matrix to obtain a supercell of given volume must be chosen so as not to reduce the number of symmetry operators.

- multiply the translational vectors by an integer i
- exploit the transformation matrix primitive $\rightarrow$ conventional cell

SUPERCEL	SUPERCON
-1 1 1	1 0 0
1 -1 1	0 1 0
1 1 -1	0 0 1

- anisotropic supercell

Running Simulations: Keywords - Vacancies (I)

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ATOMREMO: The nuclear, shell electron charges and basis set are deleted

```
ATOMREMO
1
1
ENDG
```

GHOSTS: The nuclear and shell electron charges are deleted

```
GHOSTS
1
1
ENDBS
```

Running Simulations: Keywords - Vacancies (II)

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ATOMSUBS: A new atom substitutes for a selected atom.

```
ATOMSUBS
```

```
1
```

```
1 0
```

```
ENDG
```

0 means 'ghost atom', atomic number

Running Simulations: Keywords - Vacancies (III)

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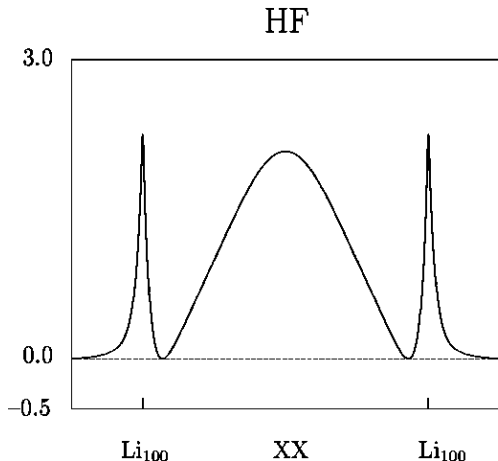
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THE EFFECTS OF THE BASIS SET

Spin density (10^{-2}
electron/bohr³)
profile along the
direction
connection the
F-centre to the
nearest neighbours
Li atoms.



Running Simulations: Keywords - Impurities (I)

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ATOMSUBS: A new atom substitutes for a selected atom.

```
ATOMSUBS
1
1 108
ENDG
```

ATOMINSE: To add new atoms in the system.

```
ATOMINSE
1
8 0.0 0.5 0.5
ENDG
```

Running Simulations: Keywords - Impurities (II)

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ATOMDISP: Selected atoms are displaced by a given vector.

```
ATOMDISP
1
1 0.0 0.0 0.1
ENDG
```

ANGSTROM, FRACTION, BOHR.

Running Simulations: Keywords - extra (I)

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MODISYMM: to break the symmetry between equivalent atoms

```
MODISYMM
1
10 1
ENDG
```

NEIGHBORS: to analyse the neighbouring atoms

```
NEIGHBORS
12
ENDG
```

Running Simulations: Keywords - extra (II)

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FRAGMENT: to investigate the perturbation of the
host crystal due to the presence of the point defect

FRAGMENT

2

4 5

ENDOPT

SPINLOCK: to control the difference between the number of
alpha and beta electrons

SPINLOCK

1 30

Running Simulations: Keywords - extra (III)

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ATOMSPIN: to define an initial spin polarised guess density for the SCF procedure

```
ATOMSPIN
```

```
1
```

```
3 1
```

```
END
```

```
ATOMSPIN
```

```
2
```

```
3 1
```

```
4 -1
```

```
END
```

EIGSHIFT: to alter the occupation of energy levels

Running Simulations:

Keywords - Label of the atom(s) (I)

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Which is the label of the atom?
Multiple - RUNs

■ PRIMITIVE CELL

```

          A           B           C           ALPHA           BETA           GAMMA
        2.84256926    2.84256926    2.84256926    60.000000    60.000000    60.000000
*****
ATOMS IN THE ASYMMETRIC UNIT      2 - ATOMS IN THE UNIT CELL:      2
      ATOM              X/A              Y/B              Z/C
*****
    1 T   9 F      0.000000000000E+00  0.000000000000E+00  0.000000000000E+00
    2 T   3 LI     -5.000000000000E-01  5.000000000000E-01  5.000000000000E-01
  
```

■ SUPERCEL/SUPERCON

■ ATOMSUBS

Running Simulations: Keywords - Label of the atom(s) (II)

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```
*****
* SUPERCELL OPTION
```

```
*****
```

EXPANSION MATRIX OF PRIMITIVE CELL

E1	2.000	0.000	0.000
E2	0.000	2.000	0.000
E3	0.000	0.000	2.000

NUMBER OF ATOMS PER SUPERCELL 16

DIRECT LATTICE VECTORS COMPONENTS (ANGSTROM)

B1	0.000	4.020	4.020
B2	4.020	0.000	4.020
B3	4.020	4.020	0.000

LATTICE PARAMETERS (ANGSTROM AND DEGREES)

A	B	C	ALPHA	BETA	GAMMA	VOLUME
5.68514	5.68514	5.68514	60.00000	60.00000	60.00000	129.929616

**** ATOMS BELONGING TO THE SUPERCELL

LABEL AT.NO.	COORDINATES (ANGSTROM AND FRACTIONARY)						
1	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	9	-2.0100	-2.0100	0.0000	0.0000	0.0000	-0.5000
3	9	-2.0100	0.0000	-2.0100	0.0000	-0.5000	0.0000
4	9	-4.0200	-2.0100	-2.0100	0.0000	-0.5000	-0.5000
5	9	0.0000	-2.0100	-2.0100	-0.5000	0.0000	0.0000
6	9	-2.0100	-4.0200	-2.0100	-0.5000	0.0000	-0.5000
7	9	-2.0100	-2.0100	-4.0200	-0.5000	-0.5000	0.0000
8	9	-4.0200	-4.0200	-4.0200	-0.5000	-0.5000	-0.5000

Running Simulations: Keywords - Symmetry (I)

ATOMSYMM:
ATOMSYMM
ENDG

```
*****
          ATOMIC SITE SYMMETRIES
*****

ATOM    1  ATOMIC NUMBER    9 - NUMBER OF SYMMOPS WHICH DO NOT MOVE THE ATOM    48
SEQUENCE NUMBERS OF THE SYMMOPS
      1   2   3   4   5   6   7   8   9   10  11  12  13  14  15
      16  17  18  19  20  21  22  23  24  25  26  27  28  29  30
      31  32  33  34  35  36  37  38  39  40  41  42  43  44  45
      46  47  48
NO EQUIVALENT ATOMS

ATOM    2  ATOMIC NUMBER    9 - NUMBER OF SYMMOPS WHICH DO NOT MOVE THE ATOM    8
SEQUENCE NUMBERS OF THE SYMMOPS
      1   2   13  14  25  26  37  38
NUMBER OF ATOMS EQUIVALENT BY SYMMETRY    5
SEQUENCE NUMBERS OF THESE ATOMS
      7   5   3   4   6
SEQUENCE NUMBERS OF THE CORRESPONDING SYMMOPS
      3   5   6   7   10
.....
```

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Running Simulations: Keywords - Symmetry (II)

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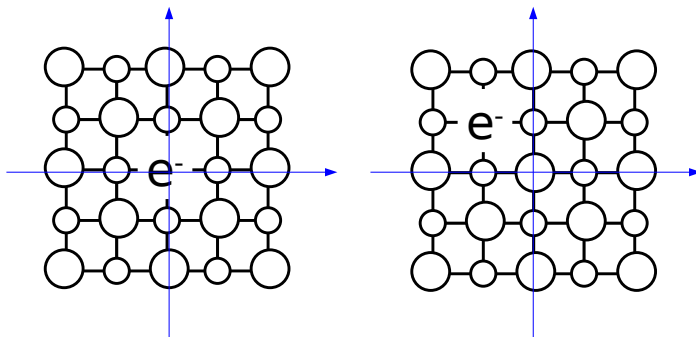
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EXPLOITATION OF POINT GROUP



Running Simulations: Keywords - Electronic Configuration

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The electronic configuration of the basis set is CRUCIAL

- ionic basis set
- atomic basis set

Defect formation energy

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NEUTRAL DEFECT

$$E_f = E_{def} - (E_{per} + \sum_i n_i \mu_i) \quad (1)$$

- E_{def} is the energy of the defective system
- E_{per} is the energy of the perfect system
- μ_i is the chemical potential of the element i

CHARGED DEFECT

$$E_f = E_{def} - E_{per} - \sum_i n_i \mu_i + q(E_F - E_V) \quad (2)$$

- q is the charge of the defect
- E_V is the valence band maximum in the perfect system
- E_F is the Fermi energy relative to E_V

Charged Defects: ENERGY OFFSET

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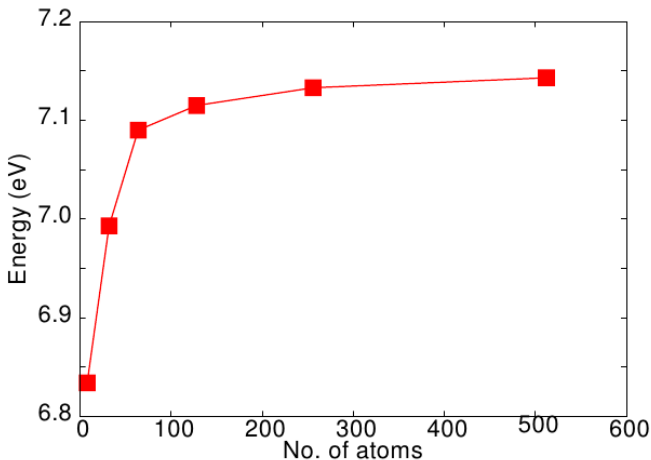
The total energy of a charged system can only be calculated to within a constant offset. The value of this constant offset depends on the average crystal potential.

HOW TO CALCULATE THE ENERGY OFFSET?

It can be obtained by calculating the change in the energy of an uncharged system when an electron is removed from it. As the system size is increased the change in energy converges to the value of the offset.

Charged Defects: Example

The change in the energy of the uncharged CuGaS_2 system when an electron is removed.



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Charged Defects: the effects of the background compensating charge

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The total energy of a periodic system with a localised charged defect treated within a finite supercell includes also term due to **defect-background** and **background-background** Coulomb interactions ⁴.

To calculate the energy of an isolated defect this term must be subtracted together with the **defect-defect** Coulomb interaction.

$$\Delta E = \frac{q^2 \alpha_M}{2\epsilon_r V^{-\frac{1}{3}}} + \frac{2\pi qQ}{3\epsilon_r V} + O(V^{-\frac{5}{3}}) \quad (3)$$

⁴C. L. Bailey et al., Journal of Physics: Conference Series, 242, 012004 (2010) doi: 10.1088/1742-6596/242/1/012004

Charged Defects: Example

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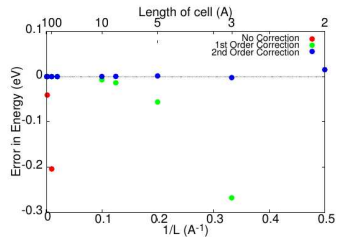
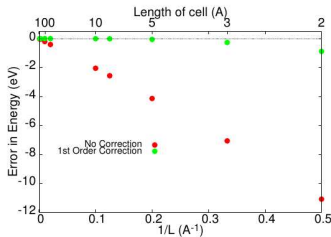
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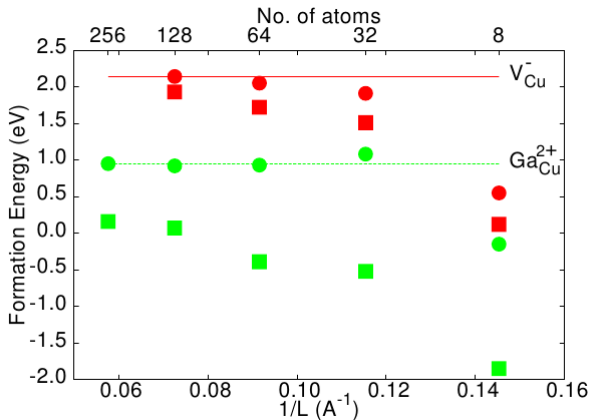
SURFACES

The difference in the calculated energy of an isolated He^+ ion and a periodic system consisting of an He^+ ion in each cell.



Charged Defects: Example

The formation energies of V_{Cu}^- (red) and Ga^{2+} (green) in $CuGaS_2$. The Cu squares denote energies before any corrections are applied and the circles denote energies after first order corrections are applied.





Running Simulations: Point Defects in Bulk

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- 1 OPTIMIZE BULK**
(lattice parameters and atomic coordinates)
- 2 GENERATE A SUPERCELL**
(concentrations of defects)
- 3 CREATE THE DEFECT**
(exploitation of the symmetry)
- 4 RELAXATION**
(optimizations of the atomic coordinates)
- 5 STUDY OF PROPERTIES**
(non-magnetic versus magnetic defects)

Running Simulations: Point Defects at Surface

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SURFACES

- 1 OPTIMIZE BULK
(lattice parameters and atomic coordinates)
- 2 **CUT A SLAB**
- 3 GENERATE A SUPERCELL
(concentrations of defects)
- 4 CREATE THE DEFECT
(exploitation of the symmetry)
(effect of layer where the defect is generated)
- 5 RELAXATION
(optimizations of the atomic coordinates)
- 6 STUDY OF PROPERTIES
(non-magnetic versus magnetic defects)



DEFECTS and ... SURFACES

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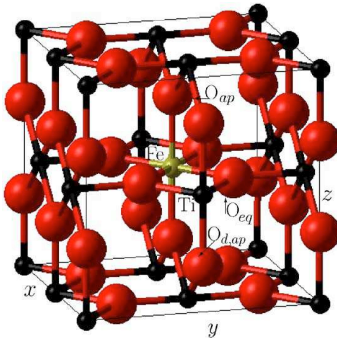
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Example

Doping titania to make a ferromagnetic semiconductor for spintronics



A rutile 24-atom supercell with an Fe atom substituted for Ti is used to simulate the dilute content of dopant ($\text{Fe}_x\text{Ti}_{1-x}\text{O}_2$, $x=0.125$).

DLV MOVIE



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Surfaces - Tasker classification

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The stability of ionic crystal surfaces

P W Tasker

Journal of Physics C: Solid State Physics, Volume 12, Number 22

Citation P W Tasker 1979 J. Phys. C: Solid State Phys. 12 4977

■ Type I:

$$q = 0$$

$$\mu_z = 0$$

■ Type II:

$$q \neq 0$$

$$\mu_z = 0$$

■ Type III:

$$q \neq 0$$

$$\mu_z \neq 0$$

α -Al₂O₃ bulk

DEFECTS and ... SURFACES

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Perfect vs
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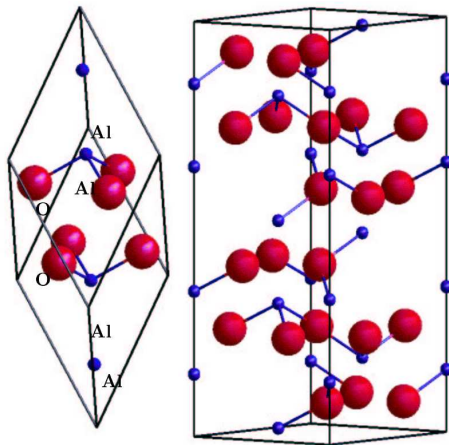
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Primitive and conventional cells.

α -Al₂O₃ planes

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Surfaces										
Plane Indices			Sequence	<i>nl</i>	<i>na</i>	<i>A</i>	<i>a</i>	<i>b</i>	Δc	<i>S</i>
C		P								
001	0001	111	Al-3O-Al	3	5	19.474	4.742	4.742	1.661	6
012	0112	110	2O-2Al-2O-2Al-2O	5	10	24.221	4.742	5.108	2.117	4
110	1120	101	O-2O-4Al-2O-O	5	10	35.416	5.108	6.970	1.846	4
100	1010	211	2O-4Al 2O-2O	3	10	61.343	4.742	12.936	0.230	4
101	1011	100	O-Al-O-Al-O	5	5	21.453	4.742	4.524	1.524	2

$\alpha\text{-Al}_2\text{O}_3$ (0001)

DEFECTS and ... SURFACES

G. Mallia

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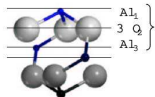
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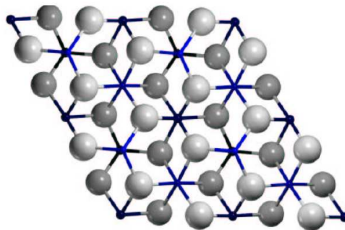
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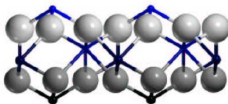
(0001)



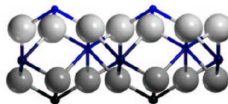
(a)



(b) TOP



(c) SIDE 1



(d) SIDE 2

α -Al₂O₃ (0001) - Relaxation

DEFECTS and ... SURFACES

G. Mallia

Perfect vs
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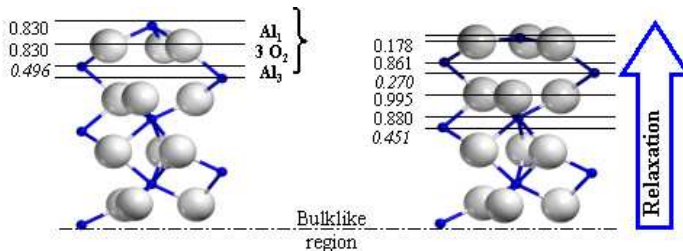
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INTERLAYER DISTANCES



Unrelaxed and Relaxed

$\alpha\text{-Al}_2\text{O}_3$ (01 $\bar{1}2$)

DEFECTS and ... SURFACES

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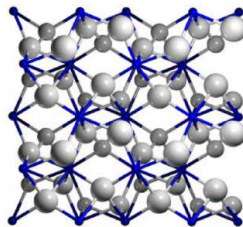
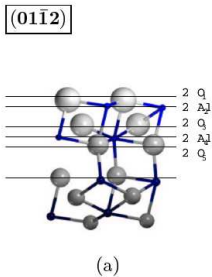
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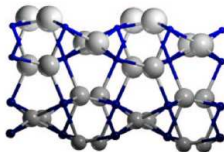
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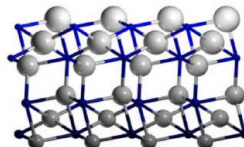
SURFACES



(b) TOP



(c) SIDE 1



(d) SIDE 2

Surfaces (01 $\bar{1}2$)

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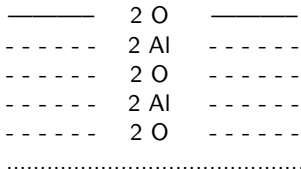
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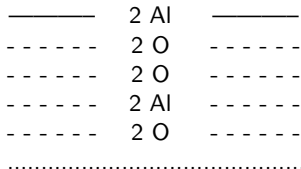
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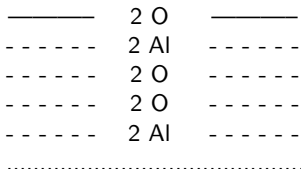
■ A, Oxygen-terminated,



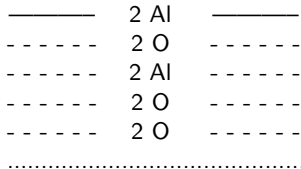
■ C, Aluminium-terminated,



■ B, Oxygen-terminated,



■ D, Aluminium-terminated,

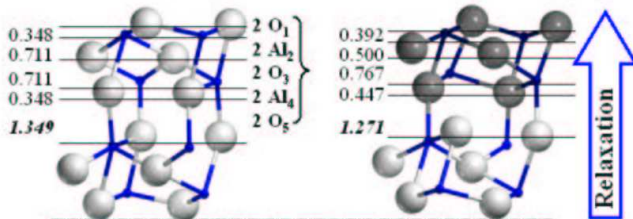


Surfaces (01 $\bar{1}$ 2) - Relaxation

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INTERLAYER DISTANCES



Bulklike region

Unrelaxed and Relaxed

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Surface formation energy

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$$E_s = \frac{E_{slab}^n - nE_{bulk}}{2A}$$

$$E'_s = \frac{E_{slab} - n(E_{slab}^n - E_{slab}^{n-1})}{2A}$$



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THANK YOU!!!